

Srujay Reddy Vangoor

AI Software Engineer · Agentic AI · LLM Platforms · RAG · FastAPI · React · AWS

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Software Engineer with 6+ years of experience designing scalable backend systems, full stack cloud native applications, and production AI platforms. Specialized in building LLM systems including retrieval augmented generation, multi agent orchestration, evaluation driven workflows, and knowledge graph augmented retrieval. Experienced in deploying distributed microservices on AWS and Azure, instrumenting production systems for observability and reliability, and modernizing legacy platforms into secure cloud native architectures.

TECHNICAL SKILLS

AI & LLM Systems: RAG Pipelines, Agentic Workflows, MCP, LangChain, LangGraph, CrewAI, Vector Search (FAISS, Pinecone), LLM Evaluation (RAGAS, LangSmith), Prompt Engineering, Hugging Face Transformers, PyTorch, Scikit learn

Backend & APIs: Python (FastAPI, Django), C# (ASP.NET Core), Node.js (Express), REST, GraphQL, WebSockets

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, OpenTelemetry, Grafana, Prometheus, Loki, Tempo

Data Systems: PostgreSQL, Redis, Kafka, DynamoDB, MS SQL Server, Snowflake, Power BI

Frontend & Mobile: React, Next.js, TypeScript, JavaScript, React Native

EXPERIENCE

AI Software Engineer

California Department of Developmental Services (DDS)

Feb 2025 – Present
Sacramento, CA (Hybrid)

- ▶ Built a conversational RAG system from scratch for internal report analysis, implementing retrieval evaluation pipelines to measure answer grounding, retrieval relevance, and response quality.
- ▶ Engineered an agentic AI workflow orchestration platform (CrewAI) with deterministic execution, schema validation, and persistent memory, reducing multi step failures from 23% to under 7% in production.
- ▶ Improved retrieval quality (hit@1 from 0.35 to 0.63) by designing a knowledge graph augmented retrieval layer with subgraph expansion, hybrid embeddings, and relevance pruning.
- ▶ Building an enterprise AI observability platform for Claude Code using OpenTelemetry and Grafana Alloy, streaming metrics, logs, and traces to Prometheus, Loki, and Tempo for centralized monitoring in Grafana.
- ▶ Designed and deployed AWS based FastAPI microservices supporting more than 200K daily API requests with sub second P99 latency for secure healthcare workflows.
- ▶ Built a full stack AI workflow intake portal using Next.js, TypeScript, Express, AWS, and Terraform, reducing submission time from 30 to 10 minutes through automated validation and review.

Software Engineer

California Department of Conservation (DOC)

Jan 2024 – Feb 2025
Sacramento, CA (Remote)

- ▶ Engineered a seismic modeling service on AWS, integrating Python (OpenQuake) with legacy PHP and Perl systems, delivering ground motion predictions in under 3 seconds per request.
- ▶ Reduced data pipeline latency by optimizing Kafka based ingestion of 10+ GB per day from earthquake monitoring stations and supporting real time analytics workloads with Apache Druid.
- ▶ Implemented identity and access management controls including RBAC, MFA, AWS Secrets Manager, and Azure Entra ID SSO to secure access across 60+ monitoring stations.
- ▶ Improved geospatial application responsiveness by optimizing Leaflet rendering and GeoServer integration for large datasets and reduced recurring downtime incidents through targeted performance fixes.

Software Application Developer Intern

Population Research Center (PRC), Sacramento State

Jan 2023 – Jun 2024
Sacramento, CA

- ▶ Built a staff management system using C#, Entity Framework, and SQL Server, optimizing stored procedures and indexing to reduce report generation and data retrieval times by 40%.
- ▶ Designed and automated health data ETL pipelines using .NET, React, and SQL Server, migrating Access and Oracle datasets into centralized systems and supporting HIPAA aligned data handling practices.

- ▶ Developed data entry tools and analytics dashboards for statewide public health studies, increasing data collection efficiency through automated validation and quality checks.
- ▶ Implemented role based access controls and server side validation across internal .NET applications to enforce secure access to sensitive public health and research data.

Software Engineer

Human Sciences Research Group (HSRG), IIIT Hyderabad

Jan 2021 – Aug 2022

Hyderabad, India

- ▶ Architected a cloud native archival and analytics platform for social media and news streams, enabling near real time analysis of over 1 million social media posts and articles during the Indian Farmers' Protests.
- ▶ Improved sentiment classification F1 score by 8% over baseline by fine tuning multilingual BERT and IndicBERT models using Hugging Face Transformers for multilingual social media analysis.
- ▶ Built 13 secure REST API endpoints using C# and ASP.NET Core for high volume data ingestion and metadata management, with React based dashboards supporting exploratory analysis and reporting.

Software Engineer

Change and Continuity in Spiti Valley, ICSSR Sponsored Project

Jul 2018 – Dec 2020

Hyderabad, India

- ▶ Built a GIS platform using Python, Django, PostgreSQL, and PostGIS to map 280+ heritage sites across 25 villages, enabling spatial search, analytics, and interactive visualization.
- ▶ Developed responsive React and offline capable React Native applications using Leaflet, reducing page load times by 60% in low bandwidth field environments.
- ▶ Automated Python ETL and spatial processing workflows integrating surveys, census, and archival datasets into a unified geospatial repository with automated validation and incremental updates.

PROJECTS

AI Software Engineering Memory – Repository Knowledge Standard

- ▶ Engineered and open sourced a lightweight repository memory standard enabling AI coding agents to persist architectural knowledge, engineering decisions, and repository conventions across development sessions.
- ▶ Designed reusable planning and review templates and validated the framework across multiple repositories to standardize AI assisted development workflows and streamline repository onboarding.

ContextServer – MCP Tool Infrastructure for AI Agents

- ▶ Built an MCP platform enabling AI agents to inspect and reason over enterprise codebases through secure filesystem access, Git history analysis, and semantic documentation retrieval.
- ▶ Architected an extensible framework with observability, execution tracing, and dynamic capability discovery, enabling new agent capabilities through declarative tool registration without changing core routing logic.

CA DMV RAG System

- ▶ Developed a RAG system using FastAPI, FAISS, and Sentence Transformers, supporting streaming responses, citation grounded answers, semantic reranking, multi document retrieval, and confidence scoring.
- ▶ Implemented evaluation and observability pipelines with distributed tracing, metrics, CI/CD automation, and automated testing to measure retrieval relevance, citation grounding, and end to end system performance.

VeryFastChat – Anonymous Real Time Chat Platform

- ▶ Engineered a cloud native real time messaging platform using Next.js, FastAPI, and Supabase, supporting instant room provisioning, persistent chat history, shareable invite links, and moderated multiuser collaboration.
- ▶ Established production readiness through Redis backed rate limiting, startup configuration validation, distributed tracing, health monitoring, GitHub Actions CI/CD, and secure deployments on Vercel and Render.

EDUCATION

Master of Science (MS) in Computer Science

California State University, Sacramento

Aug 2022 – Jan 2025

Sacramento, California

Bachelor of Technology (BTech) in Computer Science

International Institute of Information Technology, Hyderabad (IIIT-H)

Aug 2015 – Jul 2019

Hyderabad, India